

PITTSFIELD, MA, USA

WMO: 744104

Lat: 42.427N

Lon: 73.289W

Elev: 364

StdP: 97.03

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 14763

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-16.3	-25.0	0.4	-17.2	-22.4	0.5	-15.2	11.5	-5.6	10.6	-5.9	3.2	300	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	29.8	21.5	28.3	20.7	27.1	20.1	22.9	27.8	22.1	26.4	21.3	25.2	3.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.3	16.7	25.4	20.8	16.1	24.7	19.9	15.3	23.9	69.7	28.0	66.5	26.5	63.6	25.0	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.3	7.3	-23.0	32.1	2.5	1.2	-24.8	33.0	-26.2	33.7	-27.6	34.4	-29.4	35.2
			-23.1	24.6	2.6	0.9	-25.0	25.2	-26.5	25.7	-28.0	26.2	-29.9	26.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-5.4	-4.2	0.3	7.4	13.5	17.7	20.8	19.7	15.9	9.5	3.4	-1.9
	DBStd	10.22	6.31	5.61	5.51	4.81	4.33	3.41	2.68	2.96	3.80	4.57	4.78	5.26
	HDD10.0	1941	479	398	305	105	14	1	0	0	1	66	203	369
	HDD18.3	3936	737	631	560	328	161	51	8	19	92	275	447	627
	CDD10.0	1259	0	1	3	28	122	233	333	302	179	51	6	1
	CDD18.3	211	0	0	0	1	10	33	83	63	19	1	0	0
	CDH23.3	1701	0	0	0	28	134	293	658	437	144	6	0	0
	CDH26.7	363	0	0	0	6	25	65	155	84	28	1	0	0
(14) Wind	WSAvg	2.7	3.3	3.4	3.4	3.2	2.5	2.1	2.0	1.8	1.9	2.5	2.8	3.2
(15) Precipitation	PrecAvg	1142	79	63	83	79	95	112	117	112	111	112	85	96
	PrecMax	1514	177	181	158	161	184	209	287	257	227	354	160	182
	PrecMin	867	33	26	20	33	22	35	54	54	35	28	6	42
	PrecStd	172	35	30	33	32	43	53	59	55	55	68	33	33
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	14.9	20.0	27.1	29.2	31.0	32.1	31.2	29.6	23.8	18.6	15.2
		MCWB	12.1	11.1	12.2	14.9	19.0	22.3	23.2	22.7	22.0	17.4	14.2	12.5
	2%	DB	9.1	10.2	14.4	22.6	27.0	28.6	29.9	28.9	27.1	21.4	16.2	11.3
		MCWB	6.4	6.2	9.0	13.2	18.4	20.8	21.7	21.6	20.1	17.1	12.8	9.5
	5%	DB	5.2	6.6	11.2	19.3	24.5	26.5	28.4	27.3	24.8	19.3	13.8	8.4
		MCWB	3.3	3.4	6.9	11.8	16.7	19.4	21.0	20.7	19.0	15.8	10.9	6.7
	10%	DB	2.7	3.4	8.5	16.3	22.1	24.8	26.9	25.9	22.8	17.2	11.5	5.7
		MCWB	1.1	1.3	5.3	9.7	15.4	18.3	20.3	19.9	18.1	13.8	8.8	3.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.2	13.0	17.3	21.1	23.3	24.1	24.0	22.8	19.7	15.7	13.4
		MCDB	13.1	14.7	17.2	22.2	26.8	29.2	29.8	28.7	27.0	22.1	17.9	14.8
	2%	WB	6.9	6.8	10.4	15.2	19.7	21.9	23.0	22.7	21.4	18.3	13.5	9.9
		MCDB	8.3	9.6	14.0	21.3	25.1	26.6	28.2	26.9	25.3	20.3	15.4	11.4
	5%	WB	3.6	3.7	7.7	13.0	18.3	20.8	22.2	21.8	20.2	16.4	11.3	6.8
		MCDB	5.1	6.1	10.0	18.0	22.5	24.9	26.5	25.7	23.5	19.1	13.3	8.2
	10%	WB	1.2	1.5	5.2	10.8	16.9	19.6	21.4	20.9	19.0	14.1	9.1	4.0
		MCDB	2.6	3.4	8.2	14.8	20.9	23.3	25.3	24.2	21.9	16.4	11.2	5.5
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.2	9.4	11.3	11.4	10.7	10.8	10.8	11.2	9.5	9.0	7.4
		MCDBR	10.9	12.8	13.5	16.6	15.2	13.3	12.5	11.9	13.2	12.4	12.4	10.8
	5% WB	MCWBR	9.7	9.9	8.9	8.9	7.6	6.6	5.8	5.7	7.1	8.0	9.3	9.9
		MCWBR	10.7	12.2	12.3	14.6	13.0	11.4	10.6	10.2	10.8	10.6	10.8	10.2
(40) Clear-Sky Solar Irradiance	taub		0.286	0.303	0.322	0.342	0.378	0.422	0.441	0.415	0.357	0.336	0.303	0.284
		taud	2.401	2.354	2.412	2.430	2.360	2.256	2.213	2.271	2.418	2.491	2.536	2.499
	Ebn at Noon		860	899	924	927	896	853	831	843	875	854	840	837
		Edh at Noon	78	96	103	110	121	135	139	127	102	84	69	66
(44) All-Sky Solar Radiation	RadAvg		1.64	2.52	3.57	4.55	5.16	5.59	5.83	5.19	4.23	2.68	1.83	1.32
	RadStd		0.13	0.25	0.29	0.49	0.54	0.51	0.38	0.28	0.36	0.28	0.18	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+53

Nomenclature: See separate page